KEY TOPIC 4 • REVISION & PRACTICE COMPANION

Medicine in Modern Britain

c1900–present: Ideas on Causes, Diagnostic Technology, The NHS, Penicillin & Lung Cancer



Primary Visual Evidence: *Health Minister Aneurin Bevan visiting Sylvia Beckingham (first NHS patient), Park Hospital, Davyhulme, 5 July 1948* CENTRAL OFFICE OF INFORMATION ARCHIVE

PUPIL NAME: _____ **TARGET:** _____ **CLASS:** _____

KEY TOPIC 4 SPECIFICATION FRAMEWORK (1HI0/11)	LESSON DONE	QUIZ [10]
Key Topic 4.1: Ideas on Causes: Genetics, DNA & The Human Genome Project *How did Rosalind Franklin's Photograph 51, Watson & Crick, and genomic sequencing unlock hereditary disease?*	☐	☐
Key Topic 4.2: Lifestyle Factors & The Technological Revolution in Diagnosis *How did the shift to chronic lifestyle illnesses, Röntgen's X-rays, and Hounsfield's CT scanner eliminate exploratory surgery?*	☐	☐
Key Topic 4.3: Magic Bullets, High-Tech Treatments & The Birth of the NHS *How did Salvarsan 606, Prontosil, and Aneurin Bevan's 1948 NHS create free healthcare for every British citizen?*	☐	☐
Key Topic 4.4: Case Study 1: The Antibiotic Revolution: Fleming, Florey & Chain *How did Fleming's 1928 discovery, the Oxford team's purification, and American deep-tank vats deliver 2.3m doses for D-Day?*	☐	☐
Key Topic 4.5: Case Study 2: Public Health & The Fight Against Lung Cancer *How did Doll & Hill prove the tobacco link, and how did ad bans, the 2007 public smoking ban, and plain packaging end laissez-faire?*	☐	☐

Personal revision tracking ledger: ensure every specification bullet point is mastered.

LIVING TIMELINE • PART 1: MAGIC BULLETS, BLOOD GROUPS & EARLY ANTIBIOTICS (1900–1940)

Instructions: Complete the timeline sketches and notes as you master each enquiry lesson.

1901 • Karl Landsteiner: ABO Blood Groups Discovered

Key Topic 4.1

Landsteiner identifies A, B, and O blood groups, proving agglutination occurs when incompatible blood types mix. This discovery transforms blood transfusions from fatal gambles into safe, predictable surgical lifelines, foundational for modern trauma surgery.

1909 • Paul Ehrlich & Sahachiro Hata: Salvarsan 606

Key Topic 4.1

Building on dye research, Ehrlich and Hata systematically test hundreds of arsenic compounds, discovering arsphenamine (compound 606). Humanity's first synthetic chemical 'magic bullet' targets syphilis bacteria without poisoning human tissue.

1928 • Alexander Fleming: Accidental Discovery of Penicillin

Key Topic 4.2

Returning from holiday, Fleming notices a contaminated staphylococcus culture dish with a halo of destroyed bacteria around Penicillium notatum mould. Fleming publishes in 1929, though unable to purify or extract enough active substance for clinical trials.

1932 • Gerhard Domagk: Prontosil & Sulfonamide Drugs

Key Topic 4.1

Domagk demonstrates that the red leather dye Prontosil cures streptococcal blood poisoning in mice (and saves his own daughter's arm). Chemists isolate the active ingredient sulfonamide, launching the world's first mass-prescribed class of antibacterial wonder drugs.

The 20th century transformed pharmacology from toxic general poisons into targeted magic bullets.

LIVING TIMELINE • PART 2: MASS ANTIBIOTICS, THE NHS, DNA & ONCOLOGY (1940–PRESENT)

Instructions: Complete the timeline sketches and notes as you master each enquiry lesson.

1941–44 • Florey, Chain & Heatley: Penicillin Mass Production

Key Topic 4.2

Howard Florey and Ernst Chain purify penicillin at Oxford, successfully treating mice and Albert Alexander (1941). Heatley and Florey travel to Peoria, Illinois, using corn-steep liquor and deep-tank fermentation to produce 2.3 million doses for D-Day casualties.

1948 • Aneurin Bevan: Foundation of the NHS

Key Topic 4.3

On 5 July 1948, the National Health Service opens at Park Hospital, Manchester. Championed by Bevan and the 1942 Beveridge Report, it nationalises voluntary and municipal hospitals, providing universal healthcare free at the point of clinical delivery.

1953 • Watson, Crick & Franklin: DNA Double Helix Structure

Key Topic 4.4

Using Rosalind Franklin and Maurice Wilkins's X-ray crystallography (Photograph 51), Watson and Crick deduce the double-helix structure of DNA at Cambridge, unlocking hereditary genetic code and modern molecular gene therapy.

1971–2003 • CT Scanners (1971) & The Human Genome Project (2003)

Key Topic 4.4 • 4.5

Godfrey Hounsfield invents the EMI CT scanner, combining computer algorithms with X-rays to image internal organs without surgery. In 2003, the international Human Genome Project maps all 3 billion chemical base pairs in human DNA, founding precision medicine.

Why did the discovery of DNA revolutionize understanding of disease causes, yet take decades to yield treatments?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 c1900–1940s

Mendel & Early Genetics

- Mendel's pea plant laws rediscovered
- Chromosomes observed in cell nuclei
- Garrod links genes to metabolism
- Proteins wrongly assumed to carry code

FOCUS CLUE
Why did doctors assume proteins carried hereditary traits?

2 1951–1952

Franklin & Photograph 51

- Rosalind Franklin at King's College
- Expert in X-ray crystallography
- May 1952: Captures Photograph 51
- X-shaped diffraction proves double helix

FOCUS CLUE
How did Photograph 51 prove DNA's helical structure?

3 1953

Watson & Crick's Model

- Cavendish Laboratory, Cambridge
- Wilkins shares Franklin's data
- March 1953: 3D wire & metal model built
- Base pairing: A-T and C-G replication

FOCUS CLUE
How did base pairing explain hereditary replication?

4 1990–2003

Human Genome Project

- International public consortium (1990)
- Mapped all 3 billion base pairs in DNA
- Identified 20,000–25,000 human genes
- Completed 2003: genetic blueprint open

FOCUS CLUE
Why was mapping 3 billion base pairs a medical milestone?

5 2000s–Present

Screening & Treatment Gap

- Screening faulty genes (BRCA1/2, CF)
- Tailored targeted pharmacogenomics
- Treatment gap: cause known, cure hard
- CRISPR gene editing ethical debates

FOCUS CLUE
Why did understanding causes outpace finding cures?

Franklin's 62-hour X-ray exposure revealed the double helix; Watson and Crick celebrated in The Eagle pub.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why there was rapid progress in understanding the causes of disease in the period c1950 to the present. [12 marks]

Stimulus: Rosalind Franklin's Photograph 51 (1952) | The Human Genome Project (1990–2003) *(You must also use information of your own)*

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.1). In the drawing box, sketch Franklin's Photograph 51 cross pattern and Watson & Crick's double helix model. Annotate how base pairing unlocked the causes of hereditary disease! **[ ] Done**

Continue your response to Question 4 below:

Final analytical evaluation & concluding historical verdict:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** [4 MARKS • 5 MINS]

Explain one way in which scientific methods for investigating disease causes in the late twentieth century (c1950–c2000) were different from methods used in the late nineteenth century (c1870–c1900). [4 marks]

Were high-tech diagnostic scanners more vital to modern health than state lifestyle campaigns?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 **1895–1918**

Röntgen & Early X-Rays

- Wilhelm Röntgen discovers X-rays (1895)
- Passes through tissue; dense bones show
- Marie Curie runs mobile WWI X-ray cars
- Ended blind probing for bullets in war

FOCUS CLUE

How did X-rays transform surgical trauma diagnosis in WWI?

2 **1930s–1960s**

Blood Labs & Endoscopy

- Automated blood labs track chemistry
- Cellular staining detects abnormalities
- Harold Hopkins invents fiber endoscope
- Visual inspection of gut without surgery

FOCUS CLUE

Why was fiber-optic endoscopy superior to exploratory surgery?

3 **1971–1980s**

Hounsfield's CT & MRI

- Godfrey Hounsfield invents CT scan (1971)
- Rotating X-rays form 3D body slices
- Mansfield & Lauterbur develop MRI
- Radio waves image soft brain tumours

FOCUS CLUE

Why did CT scans transform internal soft-tissue diagnosis?

4 **c1950–1980s**

Epidemiological Shift

- Antibiotics conquer infectious killers
- Life expectancy rises above 75 years
- Rise of chronic non-communicable disease
- Smoking, diet, and alcohol blamed

FOCUS CLUE

What drove the shift from infectious to lifestyle diseases?

5 **1980s–Present**

State Lifestyle Campaigns

- Compulsory laws: seatbelts (1983)
- Health Act 2006: 2007 public smoking ban
- Fiscal nudge: 2018 Sugar Tax on drinks
- Education: "5 A Day" & "Change4Life"

FOCUS CLUE

Why do governments favour lifestyle prevention over treatment?

Bertha Röntgen cried "I have seen my death!" upon seeing her bones; doctors saw the dawn of non-invasive diagnosis.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

'Technological innovations in medical diagnosis were more important than public health lifestyle campaigns in improving health in Britain c1900–present.' How far do you agree? [16+4 marks]

Stimulus: Wilhelm Röntgen's X-rays (1895) | The 2007 smoking ban in public places *(You must also use information of your own)*

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.2). In the drawing box, sketch Bertha Röntgen's hand X-ray and Godfrey Hounsfield's circular CT scanner. Annotate how scanning permanently ended blind exploratory surgery! **[ ] Done**

Continue your response to Question 5/6 below:

Final analytical evaluation & concluding historical verdict:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** [4 MARKS • 5 MINS]

Explain one way in which methods of diagnosing illness in the late twentieth century (c1970–present) were different from methods used in the nineteenth century (c1800–c1900). [4 marks]

Was the establishment of the NHS in 1948 more transformative than the development of magic bullets?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 **1906–1909**

Ehrlich & Salvarsan 606

- Paul Ehrlich tests bacterial chemical dyes
- Conceives synthetic "magic bullets"
- Sahachiro Hata joins team (1909)
- Compound 606 cures syphilis in humans

FOCUS CLUE
Why was Salvarsan 606 termed the first "magic bullet"?

2 **1932–1935**

Domagk & Prontosil

- Gerhard Domagk tests red industrial dye
- 1932: Prontosil kills strep in mice
- Cures daughter's severe blood poisoning
- Active ingredient: sulfonamide drugs

FOCUS CLUE
How did Prontosil prove synthetic drugs could kill microbes?

3 **1942**

The Beveridge Report

- Sir William Beveridge social report
- Tackles "Five Giants" including Disease
- Proposes universal state social care
- Wartime collectivism spurs support

FOCUS CLUE
How did WWII unity create momentum for state healthcare?

4 **1945–1948**

Bevan & Launch of the NHS

- Aneurin Bevan appointed Health Minister
- BMA doctors resist state salaries (90%)
- Bevan lets consultants keep private beds
- 5 July 1948: NHS free at point of care

FOCUS CLUE
How did Bevan compromise to persuade resistant doctors?

5 **1950s–Present**

High-Tech Care & Costs

- Dialysis, hip replacements & transplants
- Keyhole surgery & chemotherapy units
- 1951 prescription charges divide party
- Aging population strains tax funding

FOCUS CLUE
Why has modern medical success created financial strain?

Ehrlich tested 605 failed arsenic compounds before compound 606 finally cured syphilis in 1909.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

'The establishment of the National Health Service in 1948 was the most significant turning point in medical care in the period c1900–present.' How far do you agree? [16+4 marks]

Stimulus: The National Health Service (1948) | The discovery of Salvarsan 606 (1909) *(You must also use information of your own)*

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.3). In the drawing box, sketch Ehrlich's test tube of Salvarsan 606 and the 1948 NHS information leaflet. Annotate how free healthcare abolished medical poverty! **[ ] Done**

Continue your response to Question 5/6 below:

Final analytical evaluation & concluding historical verdict:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** [4 MARKS • 5 MINS]

Explain one way in which healthcare access in Britain after 1948 was different from healthcare access in the 1930s. [4 marks]

Why did penicillin require a global war and American industrial might to become a mass-produced cure?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 1928–1929

Fleming's Discovery

- St Mary's Hospital, London (1928)
- Contaminated staphylococcus petri dish
- Penicillium notatum mould dissolves germs
- Publishes 1929; cannot chemically purify

FOCUS CLUE

Why could Fleming not turn mould into a clinical medicine?

2 1938–1940

Florey & Chain at Oxford

- Oxford team reviews Fleming's paper
- Rockefeller Foundation funds research
- Norman Heatley builds bedpan apparatus
- Freeze-drying extracts pure penicillin

FOCUS CLUE

How did Norman Heatley solve the purification challenge?

3 1940–1941

Mice Trials & Albert Alexander

- May 1940: 4 treated infected mice survive
- 1941: Policeman Albert Alexander treated
- Fatal facial septicaemia clears rapidly
- Drug runs out; patient dies; proof clear

FOCUS CLUE

Why was Alexander's trial both a success and a tragedy?

4 1941–1944

Peoria & US Mass Production

- Florey & Heatley travel to USA (1941)
- Peoria lab uses corn-steep liquor boost
- Cantaloupe mould strain yields 200x
- 10,000-gallon deep-tank vats built

FOCUS CLUE

How did US industrial fermentation enable mass production?

5 1944–Present

D-Day & Superbugs

- 2.3m doses produced for D-Day (1944)
- Wound fatality drops from 15% to 1%
- Fleming, Florey, Chain win Nobel (1945)
- Overuse drives resistant MRSA superbugs

FOCUS CLUE

Why did Fleming warn that antibiotic overuse causes resistance?

Fleming almost washed his petri dish in Lysol, but noticed a clear halo where mould destroyed golden staphylococci.

Question 4: Explain Why [12 marks • 15 mins]

Explain why penicillin was successfully developed and mass-produced in the period 1928–1945. [12 marks]

Stimulus: Alexander Fleming's discovery (1928) | The impact of the Second World War *(You must also use information of your own)*

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.4). In the drawing box, sketch Fleming's contaminated petri dish and an American 10,000-gallon deep-tank vat. Annotate how penicillin saved Allied troops on D-Day! **[ ] Done**

Continue your response to Question 4 below:

Final analytical evaluation & concluding historical verdict:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** [4 MARKS • 5 MINS]

Explain one way in which Alexander Fleming's discovery of penicillin in 1928 was similar to Edward Jenner's discovery of the smallpox vaccine in 1796. [4 marks]

Why has the modern fight against lung cancer required state coercion rather than laboratory cures?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 **1920s–1950**

The Mysterious Epidemic

- British lung cancer deaths surge 15-fold
- Doctors blame tarmac dust or motor fumes
- Tobacco companies market cigarettes
- By 1950: 65% of adult British males smoke

FOCUS CLUE
Why did early doctors fail to link smoking to lung cancer?

2 **1950–1954**

Doll & Hill's Proof

- Doll & Hill survey 20 London hospitals
- 1950 BMJ paper links smoking to tumors
- British Doctors Study tracks 40,000 GPs
- Proves heavy smokers face 50x risk

FOCUS CLUE
How did statistical epidemiology prove the causal link?

3 **1962–1971**

Royal College & Ad Bans

- 1962: Royal College report sounds alarm
- 1965: TV cigarette advertising banned
- 1971: Warning labels on tobacco packets
- Industry sponsors sports to bypass bans

FOCUS CLUE
Why did government action encounter strong tobacco lobbying?

4 **2000s**

Public Smoking Ban

- Second-hand passive smoke proven toxic
- Health Act 2006: July 2007 indoor ban
- Legal buying age raised from 16 to 18
- UK smoking falls from 65% to under 13%

FOCUS CLUE
Why was passive smoking crucial in banning public smoking?

5 **2010s–Present**

Plain Packs & Oncology

- 2016: Plain drab-olive packaging law
- Low-dose spiral CT & bronchoscopy
- Robotic VATS lobectomies & immunotherapy
- 16% 5-year survival: prevention vital

FOCUS CLUE
Why does low cancer survival prove prevention beats cure?

Doll and Hill proved smoking caused lung cancer in 1950; Doll immediately extinguished his pipe and lived to age 92.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

'Government public health legislation was the main reason for progress in combating lung cancer in the period c1950 to the present.' How far do you agree? [16+4 marks]

Stimulus: Doll and Hill's research (1950) | Technological treatments (radiotherapy, surgery, immunotherapy) | *(You must also use information of your own)*

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.5). In the drawing box, sketch a 1950s doctor cigarette advert and a 2016 olive-green plain pack. Annotate how state legislation slashed smoking from 65% to under 13%! **[ ] Done**

Continue your response to Question 5/6 below:

Final analytical evaluation & concluding historical verdict:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** [4 MARKS • 5 MINS]

Explain one way in which government intervention against lung cancer after 1965 was similar to government action in the 1875 Public Health Act. [4 marks]

Compare 1875 sewers with 2007 smoking bans: state compulsion consistently achieves greater public health impact than laissez-faire.

PUPIL NAME: _____ TARGET: _____ CURRENT GRADE: _____

ENQUIRY	ASSESSMENT COMPONENT FOCUS	PAGE	SCORE	TEACHER COMMENT & NEXT STEPS
KT4.1	**Q4:** Explain Why Rapid Progress in Causes c1950–Present	p. 5–7	/ 12	MARKING FEEDBACK • TARGETS:
	Q3: Explain One Difference (Investigating Causes: 1880s vs 1990s)	p. 7	/ 4	
KT4.2	**Q5/6:** Evaluative Essay on Diagnostic Scanners vs Lifestyle Campaigns	p. 9–11	/ 20	MARKING FEEDBACK • TARGETS:
	Q3: Explain One Difference (Diagnostic Methods: 19th c vs Late 20th c)	p. 11	/ 4	
KT4.3	**Q5/6:** Evaluative Essay on the NHS as a Medical Turning Point (1948)	p. 13–15	/ 20	MARKING FEEDBACK • TARGETS:
	Q3: Explain One Difference (Healthcare Access: 1930s vs Post-1948)	p. 15	/ 4	
KT4.4	**Q4:** Explain Why Penicillin was Mass-Produced (1928–1945)	p. 17–19	/ 12	MARKING FEEDBACK • TARGETS:
	Q3: Explain One Similarity (Fleming 1928 vs Jenner 1796)	p. 19	/ 4	
KT4.5	**Q5/6:** Evaluative Essay on Government Action & Lung Cancer	p. 21–23	/ 20	MARKING FEEDBACK • TARGETS:
	Q3: Explain One Similarity (Anti-Smoking Laws vs 1875 Public Health Act)	p. 23	/ 4	
TOTAL	**Key Topic 4 Cumulative Examination Assessment Portfolio (10 Tasks)**	p. 4–23	/ 104	**Overall Grade:** _____ Teacher Target: _____

• DIGITAL LEARNING & RETRIEVAL HUB • INTERACTIVE LESSONS & QUIZZES — Interactive Textbook • 10 Questions Per Enquiry

KT4.1	KT4.2	KT4.3	KT4.4	KT4.5
DNA & Genetics	Lifestyle & Scans	Magic Bullets & NHS	Penicillin Study	Lung Cancer Study
LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB
MASTERY QUIZ	MASTERY QUIZ	MASTERY QUIZ	MASTERY QUIZ	MASTERY QUIZ
Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]

From DNA to the NHS, state action and laboratory science revolutionized healthcare; revision guarantees your GCSE success.